

SmartCity AR

Re-imagining Cape Town's Foreshore Freeway

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Presentation Overview

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Hardware and software essentials for development and deployment.

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Core Functionalities

Key classes and methods driving the AR experience.

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Testing and Validation

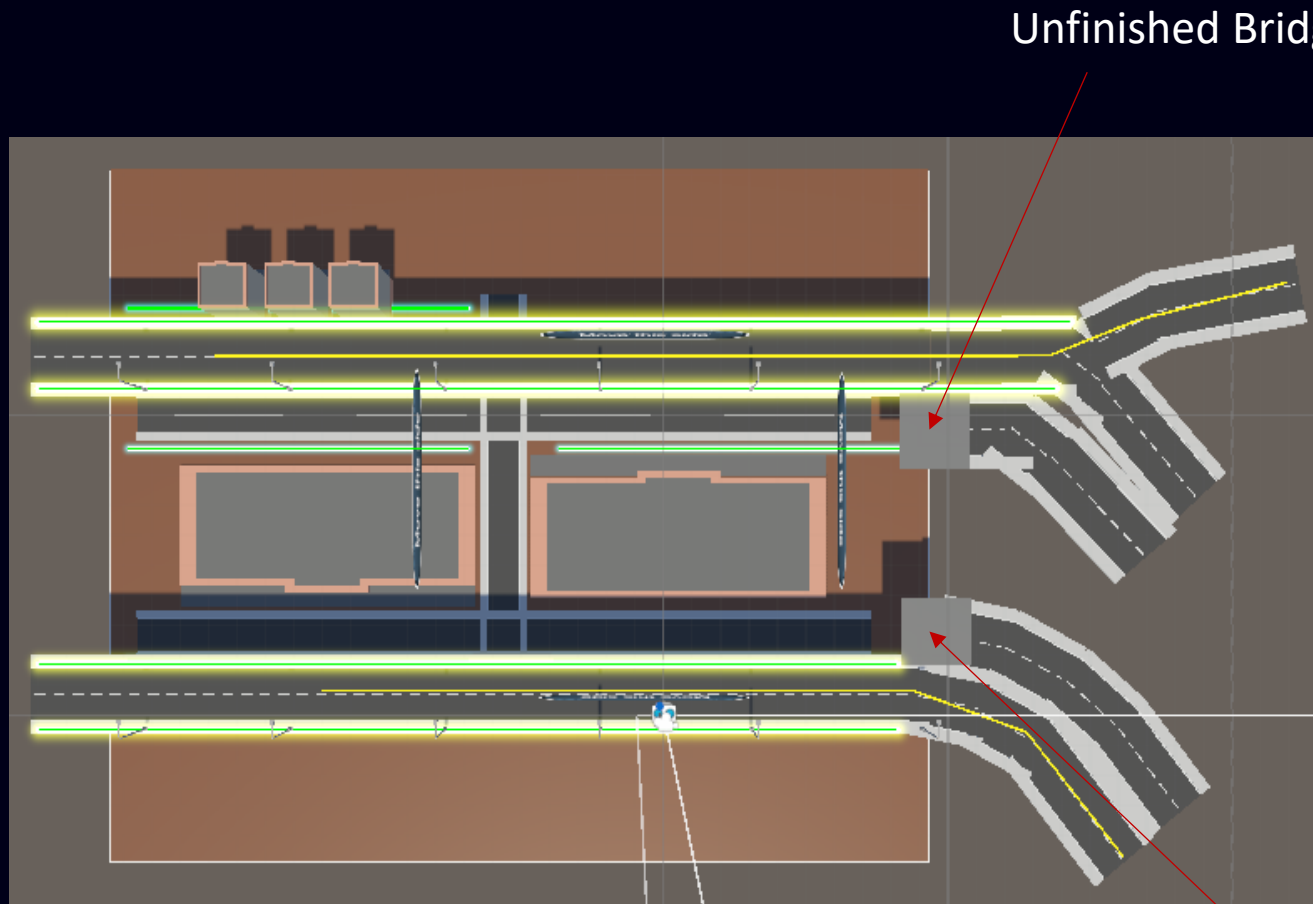
Ensuring robust performance and user experience.

05

Future Outlook

Next steps and potential enhancements for SmartCity AR.

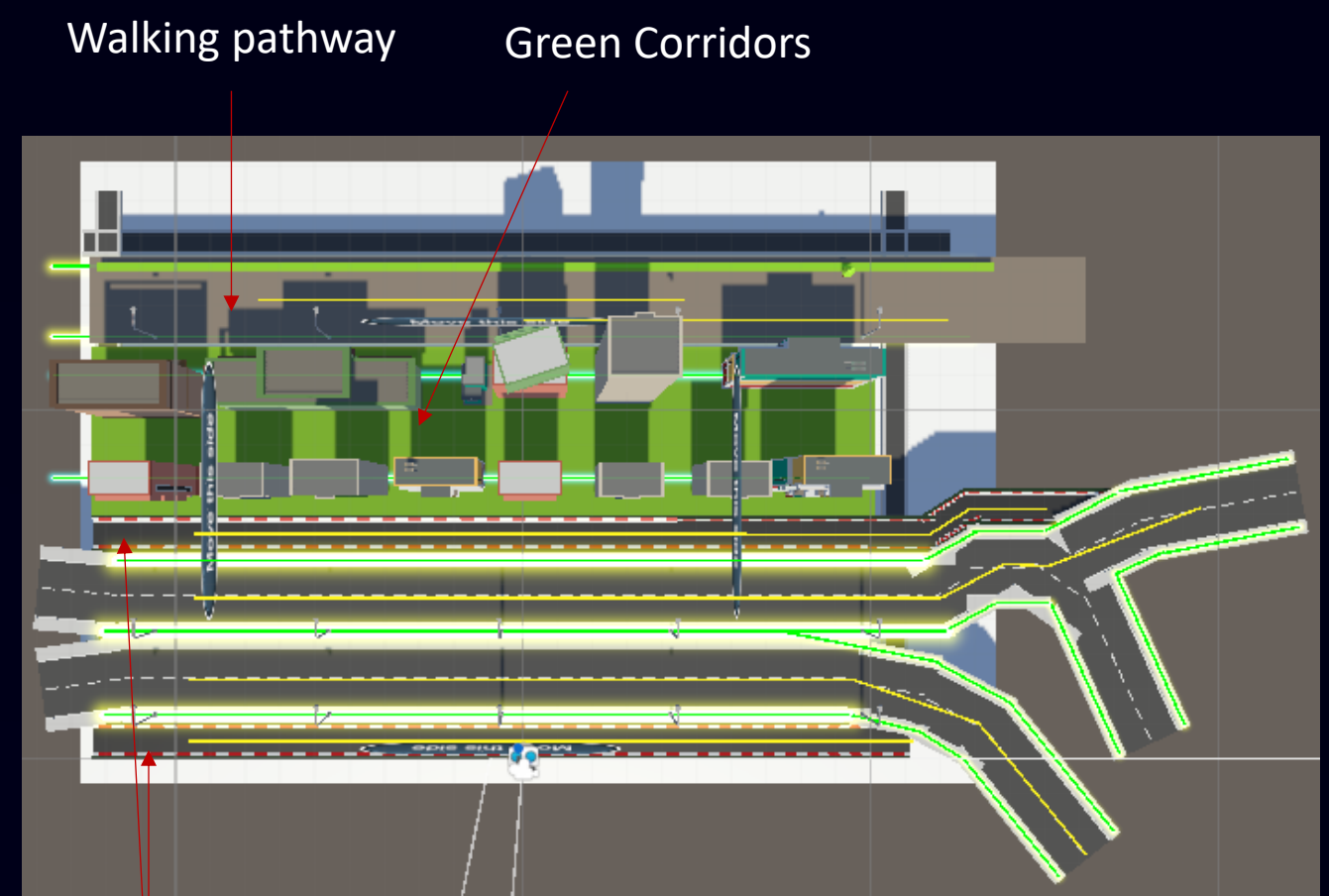
Current Situation vs Proposed Solution



Current Foreshore Freeway Highway - Unsustainable

Unfinished Bridge

Unfinished Bridge



New proposed solution - Sustainable

Walking pathway

Green Corridors

Bus Lanes



The Vision: SmartCity AR

SmartCity AR is an innovative augmented reality application designed to transform urban planning. It provides developers and city planners with a dynamic, interactive tool for visualizing and analyzing urban infrastructure in real-time.

By overlaying virtual models onto the physical environment, it enables intuitive exploration of various planning scenarios, from sustainable development to traffic management.



Hardware Requirements



Android Smartphone with ARCore Support

Essential for testing and running the AR application on a physical device, enabling accurate validation of AR interactions and surface detection. Selected for its accessibility, portability, and alignment with the project's target audience, positioning SmartCity AR as a scalable solution.



Laptop/PC for Development

Provides the computational power necessary for Unity development, 3D rendering, scripting, and building the AR application. This ensures efficient design and deployment tasks.



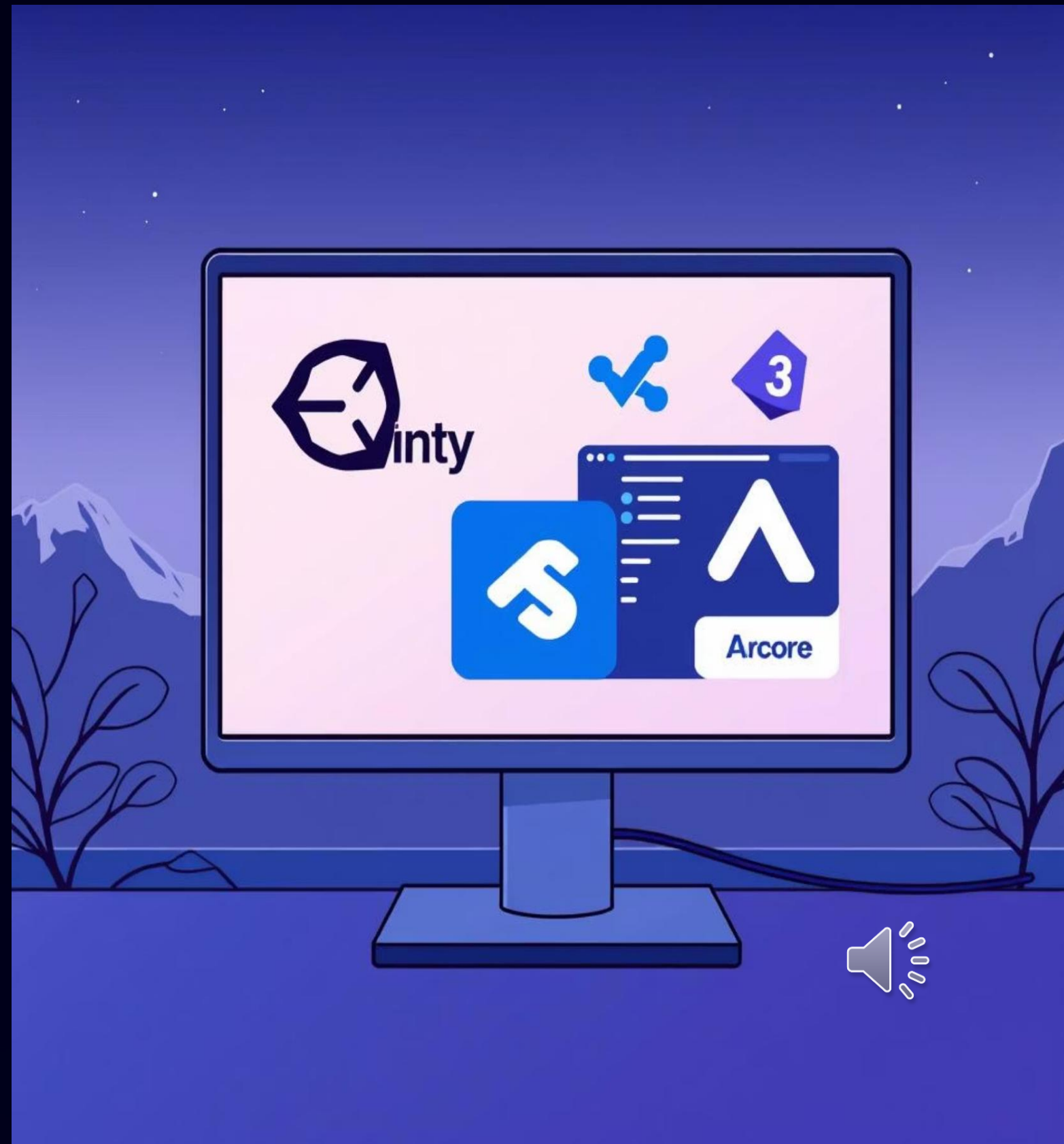
Storage and Backup Devices

An external hard drive was utilized to securely store large 3D assets, Unity project files, and AR scene backups. This practice ensures project security and robust version control.



Software Stack

- **Unity 2022.3.60f1:** The primary development platform for building, rendering, and deploying the AR environment.
- **AR Foundation 5.0:** A cross-platform framework providing core functionality for surface detection, tracking, and AR content rendering.
- **ARCore XR Plugin:** Delivers ARCore functionalities for Android devices, including plane detection, motion tracking, and light estimation.
- **Visual Studio Code 2022:** The chosen code editor for C# scripting, offering debugging, version control, and efficient project navigation.
- **Blender:** Utilized for creating and optimizing low-poly 3D models, ensuring efficient AR performance and compatibility with Unity.



Modular Code Structure

SmartCity AR's codebase is designed with modularity in mind, using a combination of classes, methods, and functions. This structure ensures each component handles specific AR features, promoting maintainability, clarity, and ease of testing.



ButtonSound
Manages audio feedback for button interactions, providing immediate auditory confirmation and enhancing user experience.



CameraMoveTrigger
Controls smooth camera transitions to focus on selected models or city areas, preventing conflicts during multiple movements.



Toggles (City, Road, Water, Electricity)
Handle visibility and state changes for different elements, allowing users to manage infrastructure layers for interactive scenarios.



ModelResetter
Repositions AR models in front of the camera and resets their orientation, ensuring consistent placement for user interactions.



RoadPath
Represents vehicle paths using waypoints and lane information, ensuring proper navigation and visual representation in the editor.

SceneResetter
Provides functionality to reset the entire AR environment by reloading the active scene, ideal for restarting simulations or testing from a clean state.

TrafficSpawner
Handles spawning vehicles along predefined paths to simulate traffic flow, utilizing object pooling for efficient performance.



Initial Testing and Validation

Limited, focused testing was conducted to verify the correct functioning of individual components within the SmartCity AR system.

- **Camera Movements:** Smooth response to user input confirmed.
- **Toggle Buttons:** Correctly show/hide city elements (sustainable/unsustainable models, roads, water, electricity).
- **Traffic Spawning:** Behaves as expected along defined paths.
- **Basic Functionality:** Model repositioning, scene resetting, and audio feedback confirmed.
- **AR Surface Detection:** Verified on supported devices, with 3D models maintaining correct scale and orientation.
- **Performance:** Informal evaluation on a single Android device showed acceptable frame rates with multiple active city elements.
- **Error Handling:** Briefly verified for missing or misconfigured objects.



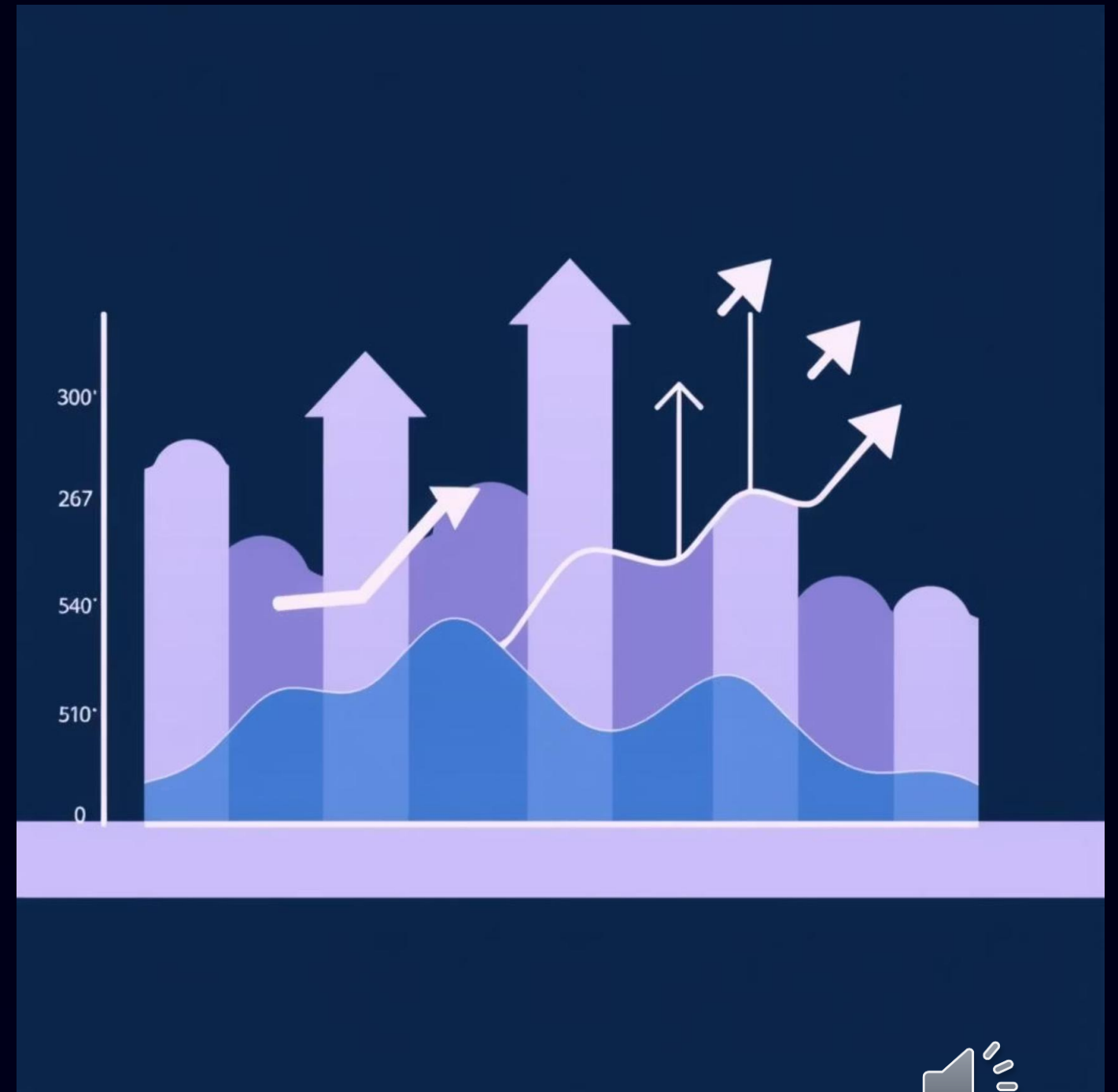
Known Limitations & Next Steps

Current Limitations:

- Testing confined to a single Android device.
- Absence of real-time traffic data integration.
- Limited stress testing for multiple simultaneous interactions.

Future Enhancements:

- **Comprehensive Integration Testing:** Full system validation.
- **Cross-Device Performance Evaluation:** Ensure broad compatibility.
- **Usability and Stress Testing:** Validate reliability under various scenarios.
- **Real-Time Data Integration:** For dynamic simulations.



Accessibility considerations, such as large tap targets, clear labels, and robust visual/audio feedback, were incorporated to accommodate a diverse user base, including those new to AR.

References

[1]	"World Population Review," 2025. [Online]. Available: https://worldpopulationreview.com/cities/south-africa/cape-town . [Accessed 2025].
[2]	J. Woosey, "IOL," [Online]. Available: https://iol.co.za/motoring/industry-news/2025-01-19-cape-town-ranked-ninth-most-congested-city-in-the-world-in-2024-inrix-global-traffic-scorecard/?utm_source . [Accessed 2025].
[3]	D. Ranasinghe, N. Kankanamge, C. D. Silva, N. Kangana, R. Mahamood and T. Yigitcanlar, "CityBuildAR: Enhancing Community Engagement in Placemaking Through Mobile Augmented Reality," MDPI, 2025.
[4]	L. Sievers, F. Othengrafen, L. Nage and E. Reinecke, "ResearchGate," Evaluating Different Levels of Detail in AR for Enhancing Citizen Participation in Urban Planning, 2024.
[5]	W. Höhl and D. Broschart, "ResearchGate," January 2015. [Online]. Available: https://www.researchgate.net/publication/283859147_Augmented_reality_in_architecture_and_urban_planning . [Accessed 2025].
[6]	DHK Architects, "Medium," 2018. [Online]. Available: https://medium.com/@dhkarchitects/citylift-dhk-425d360b3e5c . [Accessed 2025].
[7]	F. Othengrafen and L. Sievers, "ResearchGate," March 2023. [Online]. Available: https://www.researchgate.net/publication/369195465_Using_augmented_reality_in_urban_planning_processes_Sustainable_urban_transitions_through_innovative_participation . [Accessed 2025].
[8]	A. Zhezherau, "Wrike," [Online]. Available: https://www.wrike.com/agile-guide/faq/what-is-agile-iteration/ . [Accessed 2025].
[9]	"Observer Design Pattern," Geeks4Geeks, [Online]. Available: https://www.geeksforgeeks.org/observer-pattern-set-1-introduction/ . [Accessed 2025].
[10]	"State Design Pattern," GeeksforGeeks, [Online]. Available: https://www.geeksforgeeks.org/state-design-pattern/ . [Accessed 2024].
[11]	"Singleton Method Design Pattern," GeeksforGeeks, [Online]. Available: https://www.geeksforgeeks.org/singleton-design-pattern/ . [Accessed 2025].
[12]	H. Suhaizal, "The Design and Development of Augmented Reality (AR) Application for Internet Evolution Learning Topics," ResearchGate, [Online]. Available: https://www.researchgate.net/publication/369080357_The_Design_and_Development_of_Augmented_Reality_AR_Application_for_Internet_Evolution_Learning_Topics . [Accessed 2025].
[13]	"Getting started with AR Foundation," Google, [Online]. Available: https://developers.google.com/ar/develop/unity-arf/getting-started-ar-foundation . [Accessed 2025].
[14]	Aleksei, "Augmented Reality in Mobile App Design: UI/UX Challenges," Medium, [Online]. Available: https://medium.com/@Alekseidesign/augmented-reality-in-mobile-app-design-ui-ux-challenges-c1f394c11f7a . [Accessed 2025].
[15]	J. Emem, "Differences Between Monolithic and Microkernel," Baeldung, [Online]. Available: https://www.baeldung.com/cs/kernel-monolithic-vs-microkernel . [Accessed 2025].